

Sentiment Analysis and Competitive Benchmarking of Algerian Telecom Operators Using Social Media Data

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Executive Summary

This project conducts an analysis of public sentiment toward the three major Algerian telecom operators (Ooredoo, Djezzy, and Mobilis) based on Facebook posts and comments collected from their official pages. The workflow includes automated web scraping in Python, data cleaning and modeling in Power Query, and the development of an interactive Power BI dashboard for competitive comparison. An automatic sentiment analysis component was integrated using HuggingFace.

The final dashboard provides sentiment distributions, temporal evolution of public perception, operator performance benchmarking, and engagement patterns derived from both posts and user comments.

Preliminary findings indicate the following :

- Posts sentiment ranking (average score) : **Mobilis > Ooredoo > Djezzy**.
- Comments sentiment ranking (average score) : **Ooredoo > Mobilis > Djezzy**.

These insights may support strategic decision-making for marketing, communication, and customer engagement by highlighting shifts in public perception across operators.

1 Context and Objectives

1.1 Context and Problem Statement

Algerian mobile operators Ooredoo, Djezzy, and Mobilis use Facebook as a key communication channel to interact with customers, promote offers, and manage their brand image. In this context, customer posts and comments represent a rich source of information about satisfaction, recurrent problems, and expectations regarding network quality, prices, and customer service. The problem addressed in this project is to measure and better understand public perception of these three operators through a systematic analysis of posts and comments published on their official Facebook pages.

1.2 Scope of the Project

The project covers three main stages :

- **Web scraping** : automated extraction of posts and comments from the official Facebook pages of Ooredoo Algérie, Djezzy, and Mobilis, with storage in two continuously updated files, `Posts.csv` and `Comments.csv`.
- **Data transformation** : cleaning, normalization, and enrichment of the collected data using Power Query (handling dates, languages, text fields, derived variables such as engagement indicators and temporal attributes).
- **Automatic sentiment analysis** : integration of a sentiment scoring pipeline using HuggingFace and TextBlob, generating sentiment labels and scores for comments as part of the data enrichment process.
- **Data visualization** : construction of a Power BI dashboard allowing analysis of posts, sentiment analysis of comments, and competitive comparison between operators, with automatic refresh of the data model.

1.3 Objectives and Success Criteria

The general objective is to build a data visualization solution that transforms raw social media data into actionable insights on the perception of Ooredoo, Djezzy, and Mobilis.

More specifically, the project aims to :

- Develop an **automatic sentiment analysis** component (HuggingFace / TextBlob) to classify public perception expressed in posts and comments.

- Ensure clean, standardized datasets ready for analysis (consistent schema, treated missing values, derived variables).
- Provide an interactive Power BI dashboard enabling analysis of volumes, engagement, and sentiment over time, with comparison between operators.

2 Data Sources and Extraction

2.1 Data Description

The analysis is based on two main datasets designed to represent Facebook activity on the official pages of Ooredoo Algérie, Djezzy, and Mobilis.

Posts.csv This file contains information about each published post on the three official Facebook pages. The main fields include :

- **post_id** : unique identifier of the post within the dataset.
- **content** : textual content of the post.
- **post_link** : URL link to the original Facebook post.
- **nb_like** : number of “Like” reactions.
- **nb_love** : number of “Love” reactions.
- **nb_care** : number of “Care” reactions.
- **nb_wow** : number of “Wow” reactions.
- **nb_sad** : number of “Sad” reactions.
- **nb_angry** : number of “Angry” reactions.
- **nb_haha** : number of “Haha” reactions.
- **company** : name of the telecom operator owning the post.
- **date** : publication date of the post.

Comments.csv This file contains comments written by users in reaction to the posts stored in **Posts.csv**. The main fields include :

- **post_id** : identifier of the post the comment refers to.
- **user_name** : profile name of the user who wrote the comment.
- **comment** : textual content of the comment.
- **sentiment** : assigned sentiment label (positive, negative, or neutral).

These two files form the core of the analytical model : **Posts.csv** represents the publication layer, while **Comments.csv** captures the reaction layer reflecting user sentiment and engagement.

2.2 Source of the Datasets

In the initial project design, the data collection phase was intended to rely on custom web scraping of the official Facebook pages using Python scripts and suitable libraries, with automated extraction and continuous updates. However, due to time limitations and practical constraints, the web scraping component was not implemented. Instead, the team used datasets (**Posts.csv** and **Comments.csv**) already prepared and provided by the instructor, which serve as the primary source for all subsequent transformation, analysis, and visualization tasks.

3 Data Transformation and Modeling

3.1 Cleaning and Normalization Steps

The raw **Posts** and **Comments** datasets were first inspected to identify common data quality issues such as duplicated records, empty rows, and inconsistent formats. Duplicate posts and comments (same identifier and identical content) were removed to avoid over-counting engagement or sentiment, while rows containing only missing or non-informative text (e.g., blank comments) were filtered out.

Date and time values were converted to a uniform datetime type and normalized to a single reference timezone in order to enable coherent temporal analysis (daily, weekly, and monthly aggregations).

Additional normalization steps ensured that numerical fields (e.g., number of reactions such as *Like*, *Angry*, *Care*, *Haha*) were stored as numeric types. Obvious erroneous values (such as negative reaction counts) were replaced with nulls, and missing values were handled through deletion where appropriate.

These transformations resulted in clean, standardized tables ready for visualization in Power BI.

3.2 Joins and Relationships

In the data model, the **Posts** table acts as the central entity that contains information about each publication, including its identifier, date, textual content, and reaction counts. The **Comments** table stores individual user comments along with their sentiment labels.

The two tables are linked through the post identifier (**post_id** in **Posts** and **post_id** in **Comments**), establishing a one-to-many relationship : a single post may have multiple associated comments. This relational structure enables the aggregation of sentiment and engagement measures at the post and operator levels.

4 Automatic Sentiment Analysis

4.1 Method and Model Selection

Sentiment analysis was performed using pretrained models from the HuggingFace platform. More specifically, the model *CAMeL-Lab/bert-base-arabic-camelbert-da-sentiment* was used, as it is specialized in sentiment classification for Arabic dialectal text, which corresponds to the collected comments.

For each processed text, the model outputs :

- a categorical **label** (Positive / Negative / Neutral),
- an associated **confidence score**.

4.2 Processing Pipeline

The sentiment analysis pipeline implemented in Python follows these steps :

1. Load the input file *Comments.xlsx*.
2. Perform minimal cleaning : remove empty or very short comments.
3. Apply the HuggingFace classification pipeline to the **Comments** column.
4. Extract the sentiment label (**AI_Sentiment**) and confidence score (**Confidence**).
5. Save the enriched results into *Comments_Analyzed_Algeria.xlsx*.

5 Analytics and Dashboard Design

5.1 Dashboard Structure and Pages

The Power BI dashboard is composed of five interconnected pages that provide a comprehensive view of the analysis of posts and comments for Ooredoo, Djazzy, and Mobilis. A global date slicer and an operator selector enable dynamic filtering across all pages, while navigation buttons allow intuitive page transitions.

The main page displays high-level KPIs including total comments by operator (pie chart), total reactions by operator (pie chart), total post score by operator (bar chart), and average post score by operator (bar chart).



FIGURE 1 – Main page of the dashboard

5.2 Post Analytics

The publication summary page includes :

- temporal trends of average post score by operator, year, and month (line chart),
- total number of posts by operator (pie chart),
- evolution of total post score by operator, year, and month (line chart).

A detailed post table lists : **post_id**, operator, year, month, day, each reaction type (Like, Love, Care, Wow, Sad, Angry, Haha), as well as the calculated post score. The post sentiment score formula is :

$$\text{PostScore} = 2 \times \text{NbLove} + 1.7 \times \text{NbWow} + 1.5 \times \text{NbCare} + 1.2 \times \text{NbLike} + 0.7 \times \text{NbHaha} - 0.5 \times \text{NbSad} - 0.8 \times \text{NbAngry}$$

Positive reactions are weighted more heavily while negative reactions reduce the score.

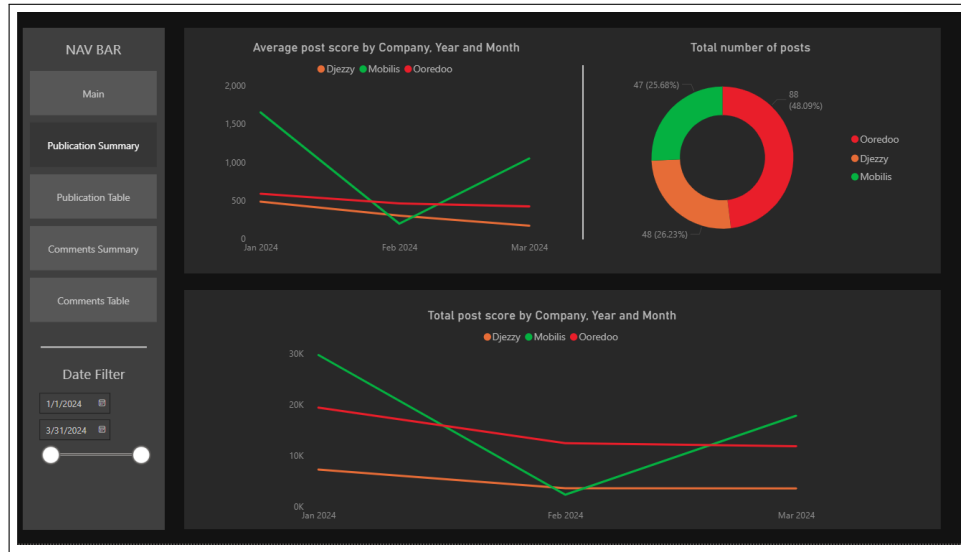


FIGURE 2 – Publication Summary

ID	Company	Year	Month	Day	Nb Angry	Nb Care	Nb Haha	Nb Like	Nb Love	Nb Sad	Nb Wow	Score
1	Djezzy	2024	January	1	76	0	14	272	45	1	1	366.60
2	Djezzy	2024	January	4	12	0	31	255	28	0	1	375.80
3	Djezzy	2024	January	10	2	2	8	257	51	0	0	417.40
4	Djezzy	2024	January	11	1	10	300	621	231	0	3	1,436.50
5	Djezzy	2024	January	12	3	4	24	288	49	0	0	464.00
6	Djezzy	2024	January	13	7	24	73	564	157	0	0	1,072.30
7	Djezzy	2024	January	14	26	5	52	438	31	0	0	610.70
8	Djezzy	2024	January	15	5	4	52	428	23	1	1	599.20
9	Djezzy	2024	January	20	2	5	9	323	37	1	0	473.30
10	Djezzy	2024	January	20	4	1	2	170	26	0	1	257.40
Total					7515	1175	2116	74512	10848	109	60	108,309.60

FIGURE 3 – Publication Table

5.3 Comment Analytics and Sentiment

The comment summary page provides :

- total comment sentiment score by operator (bar chart),
- average comment sentiment score by operator (bar chart),
- distribution of sentiment labels per operator (100% stacked bar chart),
- temporal evolution of comment sentiment by operator (line charts).

The calculated comment sentiment score is :

$$\text{CommentScore} = 1.2 \times \text{Positive} + 0.3 \times \text{Neutral} - 1 \times \text{Negative}$$

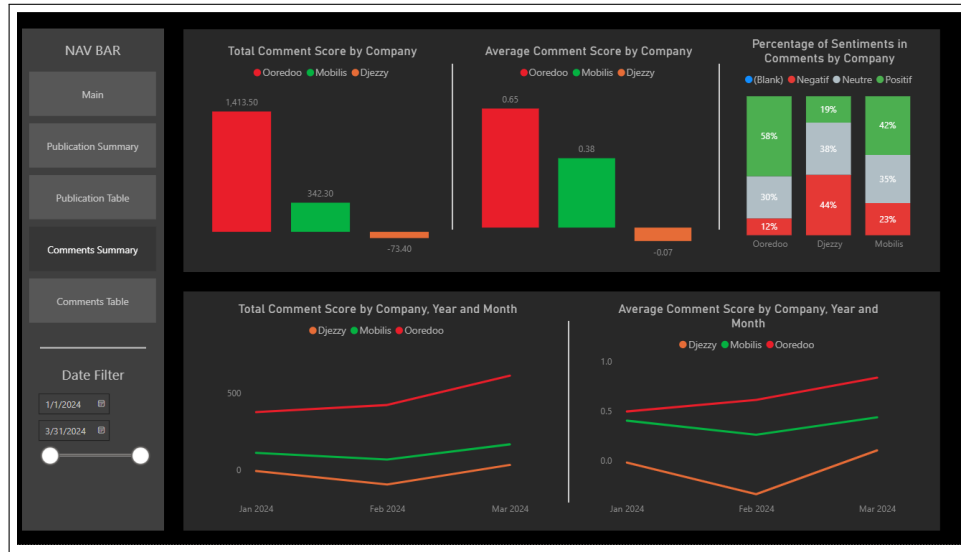


FIGURE 4 – Comments Summary

A detailed comments table displays : `post_id`, operator, date attributes, comment text, and sentiment label.

ID Post	Company	Year	Month	Day	Comments	Sentiments
1	Djezy	2024	January	1		Neutre
1	Djezy	2024	January	1	Bonne année à tous	Positif
1	Djezy	2024	January	1	Happy New year	Positif
1	Djezy	2024	January	1	Malheureusement le réseau était coupé mon 17h1 Hata dok badi wala en plus c'est en niveau d'Alger yahira les autres wilaya !! Ni excuses ni rien! Mais bon en souhaitant une amélioration cette année ndilh	Negatif
1	Djezy	2024	January	1	بشابة سنة سيئة جدا، لقد تم قطع خطي ضلّم، ومصلحة الزبائن لا ترد، عيب	Negatif
1	Djezy	2024	January	1	بطبيعة	Negatif
1	Djezy	2024	January	1	بهد ماسية نبرعو علينا ب الترتب 🍌	Neutre
1	Djezy	2024	January	1	تحيا جزيري	Positif
1	Djezy	2024	January	1	ديبولنا كش، حابة نتع كونينكيون	Neutre
1	Djezy	2024	January	1	ريزو وشبه بروج و بجي ؟	Negatif
1	Djezy	2024	January	1	ريغلو ايلكاسو نتاعكم	Negatif
1	Djezy	2024	January	1	ريغلويا رب ريزو	Negatif
1	Djezy	2024	January	1	سلام عليكم ورحمة لدا مشكلة ! طليست 100 دج و بدأت تنقص شوي من 100 دج إلى 75 دج و بعدها إلى 50 ؟؟؟؟ !!!!!! دج و بعدها إلى 25 دج !!!!!!! ان ذهبت ابوينتي تاعني !؟؟!! ان اناكم بها و لم اتصل بها و ليس لي خاصية ريني ان انا ذهبت مالي!؟؟!! !؟؟!! او صار نفس الشيء مع انا ماهلنا لاند من اسسرونا	Negatif
1	Djezy	2024	January	6	هاسنة سعيدة ورجوينا	Positif
1	Djezy	2024	January	1	عام سعيد 2024	Positif
1	Djezy	2024	January	1	علاه ديتولي 50 دج كل مرة راكم ديروهالي	Negatif
1	Djezy	2024	January	1	كاش عروض مدمم رن في 2024	Neutre
1	Djezy	2024	January	1	كل عام و انتم بخير	Positif

FIGURE 5 – Comments Table

5.4 Key Insights and Interactivity

The dashboard supports competitive benchmarking by revealing patterns in engagement and sentiment. The following insights were observed in the preliminary results (values still requiring final validation) :

- Although the total post scores of Ooredoo and Mobilis appear similar, Mobilis shows a substantially higher average post score, suggesting fewer but more impactful posts.

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- Average post score trends for Ooredoo and Djazzy indicate a decline over time, whereas Mobilis shows a decline followed by a recovery period.
 - Ooredoo exhibits the highest share of positive comments (approximately 58%), followed by Mobilis (approximately 42%).
 - Djazzy shows a notably high proportion of negative comments (approximately 44%), indicating weaker public sentiment.

These indicators enable the identification of performance gaps across operators and support decision-making for marketing communication strategies.